

7 QCs – The Seven Essential Tools for Continuous Improvement

INTRODUCTION

Quality Control (QC) is a critical aspect of manufacturing and service industries, ensuring that the products and services meet the required/established standards. Various methodologies are implemented across industries for continuous monitoring and controlling the quality of their products to achieve those standards.

One of them is the 7 QC Tools. These tools stand out as fundamental instruments for problem-solving and quality improvement. Introduced as a part of Total Quality Management (TQM), these tools were popularized by Kaoru Ishikawa and have since become essential in various industries for identifying and addressing quality related issues. Let's explore these tools one by one and see how each tool can be effectively utilized in our work along with relevant examples.



1 CHECK SHEET

A Check Sheet is a simple form used to collect and analyze data. It is one of the most intuitive QC tools and is particularly useful in tracking occurrences of specific events, such as defects or errors.

1.1 PURPOSE:

It is a structured form prepared for collecting and analyzing data. It is useful for tracking defects, occurrences, and other metrics.

1.2 HOW TO USE?

- Define the data to be collected.
- Design a simple form that facilitates data entry.
- Use the check sheet during production or service delivery to record data.
- Analyze the data for patterns or trends.

1.3 EXAMPLES:

- A plant tracking equipment breakdown could use a check sheet to log the type of malfunction, time of occurrence, and equipment involved. The resulting data would help identify patterns, such as frequent breakdowns of certain pumps or reactors.
- A retail store could use a check sheet to record customer feedback categories (e.g. long queues, stock shortages) over a month to identify the most frequent issues.

2 CAUSE AND EFFECT DIAGRAM

Also known as the Fishbone or Ishikawa Diagram (named after its creator Kaoru Ishikawa), this tool is shaped like a fish's skeleton, with the **head** representing the problem (Effect) and the **bones** representing the potential causes. It helps teams brainstorm and systematically narrow down causes of a specific issue.

2.1 PURPOSE:

Helps identify, explore and visually display the possible causes of a specific problem or quality issue.

2.2 HOW TO USE?

- Define the problem.
- Identify major categories of potential causes (Man, Machine, Material, Method, and Mother Nature in some cases also called the 4Ms/5Ms)
- Brainstorm specific causes within each category
- Analyze the diagram to prioritize causes for further investigation

2.3 EXAMPLES:

- A chemical company experiencing inconsistent product purity could use a Fishbone Diagram to examine factors like raw material variability, mixing temperatures, equipment malfunctions, or operator errors to determine the root cause.
- In a service industry, if customer complaints about long wait times are increasing, a Fishbone Diagram could help categorize potential causes such as understaffing, inefficient processes, or technological failures

3 HISTOGRAM

A histogram is a type of bar chart that displays the distribution of numerical data. It helps in understanding the frequency and patterns of variations in a dataset, making it easier to identify trends, outliers, or areas needing improvement.

3.1 PURPOSE:

A graphical representation of the distribution of numerical data. It helps in understanding the frequency and patterns of variations

3.2 HOW TO USE?

- Collect data relevant to the quality characteristics.
- Group data into ranges (bins).
- Create a bar chart with the bins on the x-axis and frequency on the y-axis.
- Analyze the histogram to identify any patterns or anomalies.

3.3 EXAMPLES:

- A chemical company might use a histogram to display the distribution of particle sizes in a batch of chemicals. If the histogram shows too many particles outside the desired size range, adjustments to milling or mixing process might be needed.
- A hotel could use a histogram to visualize the distribution of guest check-in times. If many guests are checking in outside of the standard check-in period, it might suggest operational inefficiencies.

4 PARETO CHART/ 80-20 RULE

It is named after the Italian economist Vilfredo Pareto. The chart is based on the 80/20 rule, which suggests that 80% of problems are often caused by 20% of the causes. It helps prioritize issues so that the most significant ones are addressed first.

4.1 PURPOSE:

A bar graph that displays the frequency of problems in descending order highlighting the most significant issues.

4.2 HOW TO USE?

- Identify and categorize defects or problems.
- Count the frequency of each category.
- Create a bar chart with categories on the x-axis and frequency on the y-axis.
- Analyze to focus on the **vital few** causes that require attention.

4.3 EXAMPLES:

- A Pareto Chart might show that 80% of the defects in a batch of chemicals are caused by just two problems – temperature fluctuations and incorrect raw material ratios. Addressing these would solve most of the issues.
- A customer support center could use a Pareto Chart to identify that 80% of complaints come from a small number of recurring issues, such as billing errors or slow service.

5 SCATTER PLOT

A Scatter Plot, also known as scatter diagram, visually displays the relationship between two variables. It is useful in identifying correlations (positive, negative, or no correlation) between the variable, helping to infer cause-effect relationships.

5.1 PURPOSE:

A graphical tool used to study the relationship between two variables, helping to identify correlations.

5.2 HOW TO USE?

- Determine the two variables to be analyzed.
- Collect paired data points.
- Plot the data on a scatter plot with one variable on each axis.
- Analyze the plot to identify trends or correlations.

5.3 EXAMPLES:

- A scatter plot could be used to study the relationship between ambient temperature and the viscosity of a chemical product. If a pattern is identified, the manufacturer could implement temperature controls to maintain consistent viscosity.
- A fitness center could use a scatter diagram to analyze the correlation between member attendance and promotional events. If a positive correlation is found, more events could be scheduled to boost attendance.

6 CONTROL CHART

Introduced by Walter Shewhart in the 1920s, the Control Chart (also called Shewhart Chart) is used to monitor process variability and detect whether a process is in control or exhibiting unexpected variations. It is widely used in Statistical Process Control (SPC).

6.1 PURPOSE:

A statistical tool used to monitor and control a process over time, ensuring it remains stable and predictable.

6.2 HOW TO USE?

- Collect data from the process over time
- Calculate control limits based on statistical formulas (mean, standard deviation)
- Plot the data points on the control chart.
- Analyze for variations and determine if the process is in control

6.3 EXAMPLES:

- A plant could use a control chart to monitor the pH levels of a solution during a chemical reaction. If data points begin to fall outside control limits, adjustments may need to be made to maintain product consistency.
- In a bakery, a control chart could be used to monitor the weight of bread loaves. If the weight exceeds the control limits frequently, it could indicate a problem with the dough mixing process.

7 STRATIFICATION

Stratification involves separating data into different layers or categories to reveal underlying patterns. It's often used to identify which factors contribute most to a problem by analyzing subsets of data.

7.1 PURPOSE:

Helps in separating data from different sources to identify patterns. It allows for analyzing data from different categories or sources individually to find the root cause of variations.

7.2 HOW TO USE?

- Collect data from various sources (e.g., different machines, shifts, operators).
- Separate (stratify) the data into meaningful groups (e.g., by product type, machine, or time of day).
- Analyze each stratum individually to identify trends or differences.
- Use the results to pinpoint specific causes of variations and address them.

7.3 EXAMPLES:

- A chemical plant might stratify data based in production shifts to determine why certain shifts produce a higher number of defective batches. This helps in pinpointing specific issues related to operator skills or equipment maintenance during specific shifts.
- A hospital might stratify patient wait times by department (e.g., emergency, outpatient) to identify which departments experience the longest delays and address the root cause.

CONCLUSION

The 7 QC tools - **Check Sheet, Cause and Effect Diagram, Histogram, Pareto Chart, Scatter Plot, Control Chart and Stratification** are crucial in the journey to achieving high-quality standards. These tools help organizations across various industries diagnose and resolve issues, driving efficiency, consistency, and customer satisfaction.